Physical Chemistry (II)

Lecture 16

2차 중간고사 안내 및 복습

2차 중간고사 안내

- 시간: 11월 10일(목) 오전 10:30~오후 12:00
- 장소: 화학관 315호(홀수 학번), 112호(짝수 학번)
- 계산기 불필요
- 신분증 지참

시험 범위

강의 자료 기준 10강~15강

시험 문제

총 여섯 문제

- 과제 문제 혹은 그 변형(세 문제)
  - 유도라면 중간 단계를 주지 않을 수도 있음
- 수업 시간에 배운 식 유도 및 응용(세 문제)
  - 강의 자료 기준

Overview

Hydrogen Atom

- Just H atom
- Magnetic field
- Fine structure

ψ

E

Beyond H Atom

- Identical particles
- H_2 molecule
- Helium atom

- Perturbation method
- Variational theory

Wavefunction of Hydrogen Atom

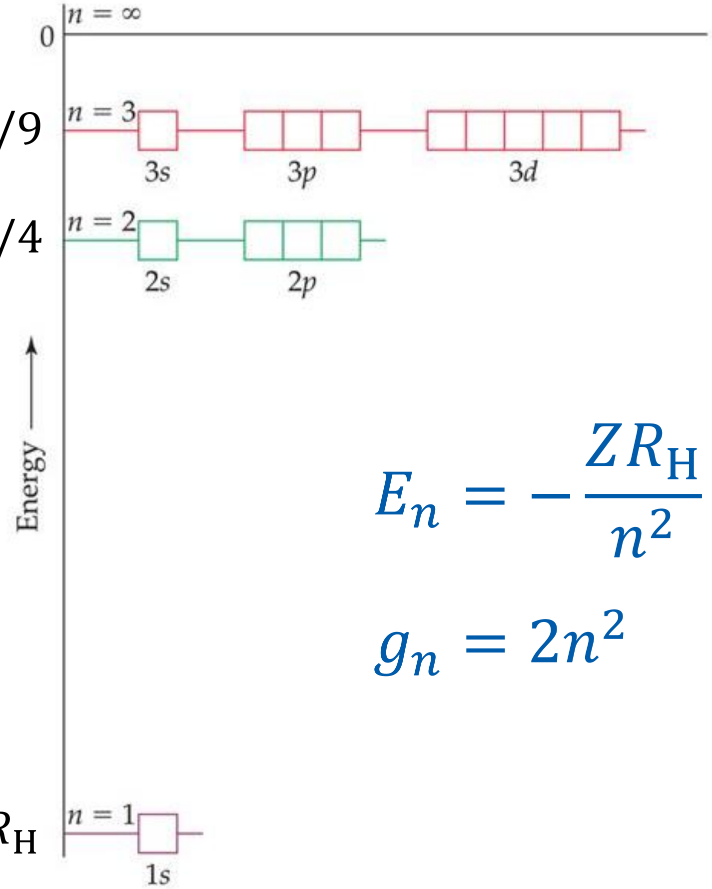
"spin" (PSet 4.3)

$$\underbrace{\psi_{nlm_l}(r,\theta,\phi)}_{R_{nl}(r)Y_l^{m_l}(\theta,\phi)}|m_s\rangle$$

$$\hat{H}_0 = -\frac{\hbar^2}{2\mu}\nabla^2 - \frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r}$$

$$\hat{H}_0\psi_{nlm_l}(r,\theta,\phi)|m_s\rangle = -\frac{Ze^2}{8\pi\varepsilon_0 a_0}\frac{1}{n^2}\psi_{nlm_l}(r,\theta,\phi)|m_s\rangle$$

Energy Levels



$$E_3 = -ZR_{\mathrm{H}}/9$$

$$E_2 = -ZR_{\mathrm{H}}/4$$

$$E_1 = -ZR_{\mathrm{H}}$$

$$E_n = -\frac{ZR_{\mathrm{H}}}{n^2}$$

$$g_n = 2n^2$$

(Image: https://slideplayer.com/slide/5959399/)

Two Sets of Quantum Numbers

One can use

$$|n\ l\ m_l\ m_s\rangle = \psi_{nlm_l}(r, \theta, \phi)|m_s\rangle$$

or

$$|n\ l\ j\ m_j\rangle = \text{linear combination of } |n\ l\ m_l\ m_s\rangle$$

Different Eigenvalues

$|n\ l\ m_l\ m_s\rangle$

$|n\ l\ j\ m_j\rangle$

$$\hat{L}^2|n\ l\ m_l\ m_s\rangle = l(l+1)\hbar^2|n\ l\ m_l\ m_s\rangle$$

$$\hat{L}_z|n\ l\ m_l\ m_s\rangle = m_l\hbar|n\ l\ m_l\ m_s\rangle$$

$$\hat{S}_z|n\ l\ m_l\ m_s\rangle = m_s\hbar|n\ l\ m_l\ m_s\rangle$$

$$\hat{L}^2|n\ l\ j\ m_j\rangle = l(l+1)\hbar^2|n\ l\ j\ m_j\rangle$$

$$\hat{J}^2|n\ l\ j\ m_j\rangle = j(j+1)\hbar^2|n\ l\ j\ m_j\rangle$$

$$\hat{J}_z|n\ l\ j\ m_j\rangle = m_j\hbar|n\ l\ j\ m_j\rangle$$

Hamiltonians

- Strong-field Zeeman effect → $|n\ l\ m_l\ m_s\rangle$

$$\hat{H} = -\frac{\hbar^2}{2\mu}\nabla^2 - \frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r} + \frac{eB}{2\mu}\left(\hat{L}_z + 2\hat{S}_z\right)$$

- Weak-field Zeeman effect → $|n\ l\ j\ m_j\rangle$

$$\hat{H} = -\frac{\hbar^2}{2\mu}\nabla^2 - \frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r} + \frac{eB}{2\mu}\left[1 + \frac{1}{2\hat{J}^2}\left(\hat{J}^2 + \hat{S}^2 - \hat{L}^2\right)\right]\hat{J}_z$$

Fine Structure

- Good quantum numbers: $|n\ l\ j\ m_j\rangle$

$$E_{\mathrm{fs}} = E_{\mathrm{rel}} + E_{\mathrm{so}} = \frac{(E_n)^2}{2mc^2}\left(3 - \frac{4n}{j+1/2}\right)$$

- Atomic term symbols denote l, s, and j

- Selection rules: $\Delta l = \pm 1;\ \Delta j = 0, \pm 1$

Identical Particles

For a wavefunction $\psi(\vec{r}_1, \vec{r}_2, \cdots, \vec{r}_N; \sigma_1, \sigma_2, \cdots, \sigma_N)$,

- Fermions: half-integer spin (electron, proton, neutron, ...)

$$\psi(\cdots, \vec{r}_i, \cdots, \vec{r}_j, \cdots; \cdots, \sigma_i, \cdots, \sigma_j, \cdots)$$
$$= -\psi(\cdots, \vec{r}_j, \cdots, \vec{r}_i, \cdots; \cdots, \sigma_j, \cdots, \sigma_i, \cdots)$$

- Bosons: integer spin (photon, helium-4 nucleus, ...)

$$\psi(\cdots, \vec{r}_i, \cdots, \vec{r}_j, \cdots; \cdots, \sigma_i, \cdots, \sigma_j, \cdots)$$
$$= \psi(\cdots, \vec{r}_j, \cdots, \vec{r}_i, \cdots; \cdots, \sigma_j, \cdots, \sigma_i, \cdots)$$

Simple Way to Construct ψ

$$\psi = \psi_{\text{spatial}}\psi_{\text{spin}}$$

For electrons, we have two options:

- antisymmetric ψ_{spatial} × symmetric ψ_{spin}
- symmetric ψ_{spatial} × antisymmetric ψ_{spin}

Spin Wavefunctions

- Triplet: symmetric
  - $|s_{\text{tot}} = 1, m_{s,\text{tot}} = 1\rangle_{\text{tot}} = |\alpha\rangle_1 |\alpha\rangle_2$
  - $|1, -1\rangle_{\text{tot}} = |\beta\rangle_1 |\beta\rangle_2$
  - $|1, 0\rangle_{\text{tot}} = \frac{1}{\sqrt{2}}(|\alpha\rangle_1 |\beta\rangle_2 + |\beta\rangle_1 |\alpha\rangle_2)$
- Singlet: antisymmetric
  - $|0, 0\rangle_{\text{tot}} = \frac{1}{\sqrt{2}}(|\alpha\rangle_1 |\beta\rangle_2 - |\beta\rangle_1 |\alpha\rangle_2)$

Helium Atom

- Simple approximation: each electron's spatial part is represented by an one-electron orbital $\phi_i(\vec{r})$

- If the two electrons are in the same orbital (*e.g.* $(1s)^2$),

$$\psi_{\text{spatial}} = \phi_i(\vec{r}_1)\phi_i(\vec{r}_2)\text{: symmetric}$$

$$\rightarrow \psi_{\text{spin}} = \frac{1}{\sqrt{2}}(|\alpha\rangle_1|\beta\rangle_2 - |\beta\rangle_1|\alpha\rangle_2)\text{: antisymmetric}$$

Helium Atom

If the two electrons are in different orbitals (*e.g.* $(1s)^1(2s)^1$), we have two options:

- antisymmetric ψ_{spatial} × symmetric ψ_{spin} (triplet)

$$\psi_{\text{spatial}} = \frac{1}{\sqrt{2}}\left[\phi_i(\vec{r}_1)\phi_j(\vec{r}_2) - \phi_j(\vec{r}_1)\phi_i(\vec{r}_2)\right]$$

- symmetric ψ_{spatial} × antisymmetric ψ_{spin} (singlet)

$$\psi_{\text{spatial}} = \frac{1}{\sqrt{2}}\left[\phi_i(\vec{r}_1)\phi_j(\vec{r}_2) + \phi_j(\vec{r}_1)\phi_i(\vec{r}_2)\right]$$

Approximation Methods

We know $\hat{H}$, but cannot obtain its eigenfunctions

1. Perturbation theory
   - If $\hat{H} = \hat{H}_0 + \hat{H}'$ and we know the eigenfunctions of $\hat{H}_0$

$$E_n^{(1)} = \int \psi_n^{(0)*} \hat{H}' \psi_n^{(0)}\, \mathrm{d}\tau = \left\langle \psi_n^{(0)} \middle| \hat{H}' \middle| \psi_n^{(0)} \right\rangle$$

2. Variational Method: ground-state energy estimation

$$E_\phi = \frac{\int \phi^* \hat{H} \phi\, \mathrm{d}\tau}{\int \phi^* \phi\, \mathrm{d}\tau} = \frac{\langle \phi | \hat{H} | \phi \rangle}{\langle \phi | \phi \rangle} \geq E_0$$